

SECTION 01 91 00.15 22

TOTAL BUILDING COMMISSIONING
08/24

AMENDMENT 009, 03-16-2026

PART 1 GENERAL

1.1 SUMMARY

The Government has hired an independent Commissioning Firm.

This Section specifies the scope of work for the entire Commissioning Team which includes the scope of work for the Lead Commissioning Specialist, Commissioning Specialists, Designers of Record, Installing Contractors, and Equipment Suppliers. Refer to paragraph COMMISSIONING TEAM for descriptions of Commissioning Team members and assigned responsibilities.

1.2 COMMISSIONING LIMITATIONS

Commissioning Requirements specified in this Section are prohibited from fulfilling requirements specified in other Sections. Requirements specified in other Sections are prohibited from fulfilling requirements specified in this Section.

1.3 SYSTEMS TO BE COMMISSIONED

Commissioning must be performed for equipment, systems, and assemblies listed under each systems category in the following paragraphs.

Specified equipment, system, and assembly counts are totals that include all commissioned buildings and all construction phases in this project. Refer to items entitled "Sample Strategy" throughout this Section that must be applied to these counts.

1.3.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category

a. Primary Systems.

- (1) Up to one steam supply system featuring one steam supply service and one condensate return system consisting of steam condensate receiver and one condensate pump, steam-to-water heat exchanger and steam pressurereducing valve.
- (2) Up to one hot water distribution system featuring up to two variable speed primary pumps, associated air separator, expansion tank, and controls.
- (3) Up to one Building Automation System.

b. Air Handling Systems.

- (1) Up to one rooftop DOAS air handling units each featuring a variable-air-volume supply air fan, a DX cooling coil, a hot water coil, air-side economizer, and packaged controls with remote VFD condensing unit.
- (2) Up to one dedicated outside air systems (DOAS) each featuring a variable-air-volume supply air fan, a variable-air-volume exhaust

air fan, a DX cooling coil, a hot water coil, air-side energy recovery, and packaged controls with remote VFD Condensing Unit.

- (3) Up to three blower coil units each featuring a direct expansion cooling coil, a hot water coil and remote condensing unit.
- (4) Up to four split system air-conditioning units each featuring packaged controls.

c. Terminal Devices.

- (1) Up to eight unit heaters (four of 8 are cabinet unit heaters) each featuring a hydronic heating coil and remote thermostat.
- (2) Up to one electric unit heaters.
- (3) Up to two exhaust air fans.
- (4) Up to two gravity ventilators with motorized dampers.

1.3.2 Plumbing Systems Category

- a. Up to two reduced pressure backflow preventers.
- b. Up to three domestic heating hot water systems each featuring up to two electric water heaters each with packaged controls, up to one recirculating pump, and up to one temperature mixing valve.
- c. Up to four emergency eyewash stations.
- d. Up to one power plant compressed air distribution system featuring packaged controls and approximately 40 drops.

1.3.3 Utility Systems Category

- a. Up to one set of whole-building energy utility metering systems.
- b. Up to one whole-building water utility metering system.

1.3.4 Electrical Type I Systems Category

- a. Power - all generator paralleling and loading systems.
- b. Power - all generator transfer switches.
- c. Power - all grounding systems.
- d. Power - all industrial control systems.
- e. Power - all main distribution panels.
- f. Power - all panelboards.
- g. Power - all step-down transformers.
- h. Power - all switchgear.
- i. Power - all transient voltage surge protective devices.

1.3.5 Electrical Type II Systems Category

- a. Energy monitoring - all individual energy monitoring of total electrical energy, HVAC systems, interior lighting, exterior lighting, and receptacle circuits.
- b. Lighting - all daylighting control systems.
- c. Lighting - all emergency egress systems, control devices, and control systems.
- d. Lighting - all exit systems, control devices, and control systems.
- e. Lighting - all exterior systems, control devices, and control systems.
- f. Lighting - all interior systems, control devices, and control systems.
- g. Power systems - all automatic receptacles controls.

1.3.6 Building Envelope Systems, Components, and Assemblies Category

- a. All building envelope systems including curtain walls, exterior penetrations, exterior air barrier assemblies, exterior windows, exterior doors, and skylights.
- b. All roofing including low sloped roofs and steep sloped roofs.

1.4 COMPLEX SYSTEMS

Complex systems are defined as the following systems categories as listed under paragraph SYSTEMS TO BE COMMISSIONED with appropriate qualifiers and are subject to more stringent requirements as specified in this Section.

- a. All equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category."
- b. All equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" featuring on-board factory packaged controls such as domestic heating hot water systems, domestic solar heating hot water systems, and service water pressure-booster pump systems. Refer to the following paragraph for definition of equipment featuring on-board factory packaged controls.
- c. All equipment, systems, and assemblies listed under item entitled "Utility Systems Category."

Equipment featuring on-board factory packaged controls is defined as equipment that is a complex system as defined in paragraph COMPLEX SYSTEMS which features a controller issuing sophisticated logic-based commands to its internal components to accomplish the intent of Plans and Specifications. This equipment may function as either a stand-alone piece of commissioned equipment or may function as part of a commissioned system. This controller may be equipment mounted or may be remote mounted and may or may not interface with a building control system. Examples of equipment meeting and not meeting this definition are provided in the following paragraphs.

- a. Equipment meeting this definition includes chillers, boilers provided with or without master controllers, packaged pumping systems,

dedicated outside air systems, packaged rooftop units, computer room air conditioners, and air terminal units.

- b. Equipment not meeting this definition includes unitary air-conditioners, packaged terminal air-conditioners, and heat pumps.
- c. Equipment not meeting this definition, but requiring demonstration of minimum safe motor speed either prior to or during Functional Performance Testing includes variable frequency drives.

1.5 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019; Errata 1 2019; Errata 2-5 2020; Addenda BY-CP 2020; Addenda AF-DB 2020; Addenda A-G 2020; Addenda F-Y 2021; Errata 6-8 2021; Interpretation 1-4 2020; Interpretation 5-8 2021 Addenda AS-AQ2022) Energy Standard for Buildings Except Low-Rise Residential Buildings

1.6 SUBMITTALS

Commissioning Submittals are not required to be submitted by the Installing Contractor. Exchange of information between the Commissioning Team specified in this Section must still occur.

1.7 COMMISSIONING TEAM

The Commissioning Team is defined as the entity that must perform all Commissioning Activities specified in this Section for systems listed under paragraph SYSTEMS TO BE COMMISSIONED. The Commissioning Team is composed of the Lead Commissioning Specialist, Commissioning Specialists, Designers of Record, Installing Contractors, Equipment Suppliers, and the Contracting Officer's Technical Representative.

1.7.1 Lead Commissioning Specialist

The Lead Commissioning Specialist (Cx) is defined as the entity that must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, coordinating all aspects of the Commissioning Process; managing all Commissioning Specialists; and submitting all Commissioning Plans, schedules, reports, and documentation directly to the Contracting Officer's Technical Representative as modified in this Section.

1.7.2 Commissioning Specialists

Commissioning Specialists are defined as entities that must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, performing reviews, witnessing that all system components have been installed, that each control device operates, that each equipment piece operates, that complete systems operate, and that

interactive systems operate in accordance with Plans and Specifications.

Commissioning Specialists must perform assigned responsibilities for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

1.7.3 Designers of Record

Designers of Record are defined as entities that must act as design professionals responsible for design of their respective system that is subject to Commissioning. Designers of Record must perform all Commissioning Responsibilities assigned in this Section.

1.7.4 Installing Contractors

Installing Contractors are defined as the General Contractor's Superintendent or Project Manager; Quality Control Representative; and Mechanical, Controls, Plumbing, Electrical, Test and Balance, Fire Protection, and Building Envelope Sub-Contractors who provide, install, or construct systems in whole or in part for this contract. The General Contractor must provide a Lead Commissioning Representative to manage all Sub-Contractor Commissioning Representatives and Equipment Suppliers to accomplish all Commissioning Responsibilities assigned in this Section. Each Sub-Contractor must provide a Commissioning Representative to coordinate and perform all Commissioning Responsibilities for commissioned equipment, systems, and assemblies provided as part of their respective contract as directed by the General Contractor. Submission of Commissioning related certifications or qualifications for the Lead Commissioning Representative and Installing Contractor Representatives is not required.

Installing Contractors must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, demonstrating performance of all commissioned equipment, systems, and assemblies provided as part of their respective contract and must provide all necessary materials, services, and labor required for successful demonstration as modified in this Section.

1.7.5 Equipment Suppliers

Equipment Suppliers (ES) are defined as factory authorized start-up technicians or factory representatives knowledgeable with start-up, demonstration, and testing of commissioned equipment featuring on-board factory packaged controls provided as part of their respective contract. Each Equipment Supplier must coordinate and perform all Commissioning Responsibilities assigned in this Section as directed by the General Contractor. Submission of Commissioning related certifications or qualifications for Equipment Suppliers is not required. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls's.

Equipment Suppliers must perform all Commissioning Responsibilities assigned in this Section that includes, but is not limited to, demonstrating performance of all commissioned equipment provided as part of their respective contract and must provide all necessary materials, services, and labor required for successful demonstration of all

commissioned equipment provided as part of their respective contract as modified in this Section.

1.7.6 Installing Contractors and Equipment Suppliers Coordination

Installing Contractors and Equipment Suppliers must coordinate and share demonstration and testing responsibilities specified in this Section for commissioned equipment featuring on-board factory packaged controls. Installing Contractors and Equipment Suppliers must simultaneously perform all aspects of demonstration and testing for these types of systems. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls.

1.7.7 Contracting Officer's Technical Representative

The Contracting Officer's Technical Representative (COTR) represents the Government for all elements of the Commissioning Process. The COTR will serve as the single point of contact with the Lead Commissioning Specialist, will receive all Commissioning Submittals, may attend meetings, and may, but is not required to, witness field inspections and testing specified in this Section.

1.8 COMMUNICATIONS

1.8.1 Communications with the Contracting Officer's Technical Representative

Communication with the Contracting Officer's Technical Representative must be through the Lead Commissioning Specialist who acts as the single direct point of contact for all aspects of the Commissioning Process.

1.8.2 Communications between the Commissioning Team

Communications between the Commissioning Team including, but not limited to, verbal communications, schedule information, and exchange of information specified in this Section, must occur directly between the Commissioning Team without requiring interaction of the Contracting Officer's Technical Representative.

1.9 NUMBER OF BUILDINGS AND CONSTRUCTION PHASING

1.9.1 Number Of Buildings

This project includes one commissioned building.

- a. Construction Phase Site Observations specified in this Section are inclusive of all commissioned buildings in this project. Refer to item entitled "Construction Phase Site Observations" for additional requirements.
- b. Construction Phase Functional Performance Testing must be performed during a single continuous occurrence inclusive of all commissioned buildings for each systems category until completion of testing of all systems listed under respective systems category. Refer to item entitled "Construction Phase Functional Performance Testing" for additional requirements.
- c. Post-Construction Phase Site Observations specified are inclusive of all commissioned buildings in this project. Refer to item entitled

"Post-Construction Phase Site Observations" for additional requirements.

1.9.2 Construction Phasing

This project's construction is not phased.

1.10 ASHRAE STANDARD 90.1 COMMISSIONING COMPLIANCE

ASHRAE Standard 90.1 Commissioning compliance is required for this project.

1.10.1 General Requirements

All Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Commissioning Activities listed in references contained in ASHRAE 90.1 - IP's paragraphs as modified in this Section, and Commissioning Activities specified in this Section must be performed. All Commissioning Activities that are referred to as "optional activities," "guidance," "should activities," "recommendations," "suggested practices," "informative information," or listed as similar designations are mandatory regardless of any qualifiers.

1.10.2 Commissioning Team Responsibilities

The Commissioning Team must, at a minimum, perform all Commissioning Activities specified in the preceding item entitled "General Requirements" and must perform all Commissioning Activities specified in this Section. This Section takes precedence should there be a conflict between requirements listed in this Section and requirements listed in ASHRAE 90.1 - IP or listed in any of its references.

1.11 SUSTAINABILITY THIRD PARTY CERTIFICATION (TPC) COMMISSIONING COMPLIANCE

Sustainability Third Party Certification (TPC) is not required for this project.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 CONSTRUCTION PHASE

3.1.1 Construction Phase ASHRAE Standard 90.1 Commissioning Compliance

3.1.1.1 General Requirements

Activities must include, but not be limited to, Construction Phase Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Construction Phase Commissioning Activities listed in references contained in ASHRAE 90.1 - IP as modified in this Section, and Construction Phase Commissioning Activities specified in this Section.

3.1.1.2 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

The Lead Commissioning Specialist and Commissioning Specialists must, at a minimum, perform all activities specified under item entitled "Construction Phase ASHRAE Standard 90.1 Commissioning Compliance."

3.1.2 Construction Phase Sequencing

3.1.2.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems

Heating, ventilation, and air-conditioning (HVAC) systems activities specified in the following paragraphs must be performed as applicable prior to performing Functional Performance Testing of systems listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

- a. All equipment, systems, and assemblies have been completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with Plans and Specifications.
- b. All Ductwork Air Leakage Testing (DALT) field work, Test and Balance (TAB) field work, Pre-Final TAB Reports, and related prerequisites have been completed, submitted, and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.
- c. All inspections, measurements, and testing of building envelope and roofing systems specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES have been completed, submitted, and approved.
- d. All pressure testing, infrared thermography testing, and fog testing of building envelope and roofing systems specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING have been completed, submitted, and approved.
- e. All Pre-Functional Checks have been completed, submitted, and approved.
- f. Certified TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC has been prepared, submitted, and approved.
- g. Certified Certificate of Readiness has been completed, submitted, and approved.

All completed Final Functional Performance Test Checklists for commissioned equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must be submitted and approved prior to the Government performing any Performance Verification Testing specified in Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC.

3.1.2.2 Electrical Systems

Electrical systems activities specified in the following paragraphs must be performed as applicable prior to performing Functional Performance Testing of systems listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

- a. All electrical, power generation, and lighting equipment, systems, and assemblies have been completed, calibrated, tested, and operate in accordance with Plans and Specifications.
- b. Building Envelope has been enclosed according to contract documents and final construction has been completed.
- c. All ceiling tiles, floor coverings, and window coverings have been put in place.
- d. All lamps have completed a minimum 100-hour burn-in period.
- e. All furniture has been put in place.
- f. Certified Certificate of Readiness has been completed, submitted, and approved.

3.1.3 Construction Phase Scheduling

3.1.3.1 General Requirements

Commissioning Activities and other related activities for inclusion into the Project Construction Schedule must include, but not be limited to, activities specified in the following paragraphs.

- a. Perform Construction Phase Commissioning Coordination meeting.
- b. Submit and approve completed Pre-Functional Checklists.
- c. Submit and approve proposed Functional Performance Test Checklists.
- d. Submit and approve final Functional Performance Test Checklists.
- e. Perform Construction Phase Site Observations.
- f. Perform inspections, measurements, and testing of building envelope and roofing systems specified in Section 07 27 10 BUILDING AIR BARRIER SYSTEM.
- g. Perform pressure testing, infrared thermography testing, and fog testing of building envelope and roofing systems specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.
- h. Submit and approve Certified TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING HVAC.
- i. Submit and approve Certified Certificate of Readiness.
- j. Witness Construction Phase Functional Performance Testing.
- k. Correct deficiencies discovered during Construction Phase Functional Performance Testing.
- l. Re-witness Construction Phase Functional Performance Testing.
- m. Government Witness of Performance Verification Testing.
- n. Correct deficiencies discovered during Performance Verification Testing.

- o. Government Re-witness of Performance Verification Testing.
- p. Provide all Systems Manuals contents.
- q. Submit and approve completed Systems Manuals.
- r. Perform training.
- s. Submit and approve Final Commissioning Report.
- t. Perform Post-Construction Phase Site Observations.
- u. Witness Post-Construction Phase Functional Performance Testing if all requirements and activities for respective system could not be fully or successfully completed during the Construction Phase.
- v. Correct deficiencies discovered during Post-Construction Phase Functional Performance Testing.
- w. Re-witness Post-Construction Phase Functional Performance Testing.

3.1.3.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must request a current Project Construction Schedule from Installing Contractors no later than at the Construction Phase Commissioning Coordination Meeting and must prepare and provide to Installing Contractors a list of Commissioning Activities for inclusion into the Project Construction Schedule. The Lead Commissioning Specialist must associate each Commissioning Activity to a single construction activity such that if the date of the construction activity changes, the associated Commissioning Activity changes accordingly. List of Commissioning Activities for inclusion into the Project Construction Schedule is not required to be submitted to the Government.

3.1.3.3 Installing Contractors Responsibilities

Installing Contractors must provide a current Project Construction Schedule to the Lead Commissioning Specialist upon receiving request. Installing Contractors must incorporate the list of Commissioning Activities into the Project Construction Schedule and must link each Commissioning Activity to the associated construction activity identified by the Lead Commissioning Specialist such that if the date of the construction activity changes, the associated Commissioning Activity changes accordingly.

Installing Contractors must provide updated Project Construction Schedules to the Lead Commissioning Specialist at least on a monthly basis.

3.1.4 Construction Phase Commissioning Coordination Meeting

3.1.4.1 General Requirements

A Construction Phase Commissioning Coordination meeting must be conducted no later than 90 calendar days after award of construction contract to discuss topics that include, but not be limited to, the Commissioning Process, contract requirements, lines of communication, roles and responsibilities, Commissioning Activities for inclusion into the Project Construction Schedule, documentation requirements, Inspections and

Testing, and logistics specified in this Section. Meeting must be conducted by conference call.

3.1.4.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, conduct meeting and prepare meeting minutes complete with Attendance Roster.

3.1.4.3 Installing Contractors and Others Responsibilities

Installing Contractors must attend this meeting. The Contracting Officer's Technical Representative, a User Representative, and a Public Works Department Representative may attend this meeting.

3.1.5 Construction Phase Commissioning Progress Meetings

3.1.5.1 General Requirements

Construction Phase Commissioning Progress meetings must be conducted to discuss topics that include, but not be limited to, unresolved Commissioning Issues, status of items specified to be provided to the Lead Commissioning Specialist, and upcoming Commissioning Activities. Number of Construction Phase Commissioning Progress meetings must be conducted for each systems category as specified in the following paragraphs. Meetings must be conducted by conference call.

- a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: Six meetings.
- b. Plumbing Systems Category: Two meetings.
- c. Utility Systems Category: No meetings.
- d. Electrical Type I Systems Category: No meetings.
- e. Electrical Type II Systems Category: Two meetings.
- f. Building Envelope Systems, Components, and Assemblies Category: Two meetings.

Construction Phase Commissioning Progress meetings are prohibited from being combined with conducting Construction Phase Commissioning Coordination meeting, conducting Site Observations, or witnessing Functional Performance Testing.

3.1.5.2 Commissioning Specialists Responsibilities

Each Commissioning Specialist must, at a minimum, conduct meetings and prepare meeting minutes complete with Attendance Roster for systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for systems categories assigned to other Commissioning Specialists.

3.1.5.3 Installing Contractors and Others Responsibilities

Installing Contractors must attend these meetings. The Contracting Officer's Technical Representative, a User Representative, and a Public Works Department Representative may attend these meetings.

3.1.6 Construction Phase Interim Commissioning Plan

3.1.6.1 General Requirements

The Construction Phase Interim Commissioning Plan must be prepared and must be an update to the Design Phase Commissioning Plan and contents must include, but not be limited to, contents specified in the following paragraphs.

- a. All contents specified for the Design Phase Commissioning Plan.
- b. Basis of Design.
- c. Contact information for the entire Commissioning Team.

3.1.6.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, prepare and submit the **Construction Phase Interim Commissioning Plan** no later than 30 calendar days after the Construction Phase Commissioning Coordination meeting and no later than 14 calendar days prior to performing construction of the building envelope.

3.1.7 Construction Phase Final Commissioning Plan

3.1.7.1 General Requirements

The Construction Phase Final Commissioning Plan must be prepared and must be an update to the Construction Phase Interim Commissioning Plan and contents must include, but not be limited to, contents specified in the following paragraphs.

- a. All contents specified for the Construction Phase Interim Commissioning Plan.
- b. All Pre-Functional Checklists.
- c. All approved final Functional Performance Test Checklists.

3.1.7.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

- a. Prepare and submit the **Construction Phase Final Commissioning Plan** no later than 30 calendar days prior to performing any Functional Performance Testing.

3.1.8 Construction Phase Issues And Resolution Log

3.1.8.1 General Requirements

The Construction Phase Issues and Resolution Log must be prepared and maintained to track all Commissioning Issues identified by any member of the Commissioning Team. The Construction Phase Issues and Resolution Log must be prepared in an electronic format that can be used throughout the Construction Phase such that it can be provided to the Government upon request. Construction Phase Issues and Resolution Log contents must include, but not be limited to, contents specified in the following

paragraphs.

- a. Project title and location.
- b. Lead Commissioning Specialist's contact information.
- c. Issue number.
- d. Date issue was discovered.
- e. Name of Commissioning Team member identifying the issue.
- f. Commissioning Team member responsible for resolving the issue.
- g. Current status.
- h. Issue description.
- i. Pictures of item if available and appropriate.
- j. Effects of issue on project or building operation.
- k. Possible cause of issue.
- l. Proposed resolution.
- m. Final resolution and final resolution date.
- n. Approvals of issue resolution, including approver's name.

3.1.8.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must prepare and maintain the **Construction Phase Issues And Resolution Log** and must submit one electronic copy no later than seven calendar days after the addition of an issue or a change in status of an issue.

3.1.9 Pre-Functional Checklists

3.1.9.1 General Requirements

Pre-Functional Checklists must be prepared for all commissioned equipment, systems, and assemblies that are subject to functional performance testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Pre-Functional Checklist contents must include, but not be limited to, contents for physical inspections that demonstrate construction, installation and start-up of equipment, systems, and assemblies is complete. Refer to item entitled "Pre-Functional Checklists" in item entitled "Inspections and Testing" for more information.

Download example Pre-Functional Checklists for Section 01 91 00.15 TOTAL BUILDING COMMISSIONING at the following location: <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Checklists must contain the same level of detail shown in examples, but are not required to match the format of examples.

3.1.9.2 Lead Commissioning Specialist Responsibilities

Lead Commissioning Specialist must provide blank Pre-Functional Checklists to Installing Contractors. Blank Pre-Functional Checklists are not required to be submitted to the Government.

3.1.9.3 Commissioning Specialists Responsibilities

Commissioning Specialists must prepare Pre-Functional Checklists for all equipment, systems, and assemblies listed under systems categories assigned to them in this Section that are subject to functional performance testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.10 Functional Performance Test (FPT) Checklists

Functional Performance Test (FPT) Checklists must be prepared for all commissioned equipment, systems, and assemblies demonstrating performance is in accordance with Plans and Specifications as specified in the following paragraphs. A single final FPT Checklist must be prepared for each equipment piece, system, and assembly except that a single final FPT Checklist may be prepared for use with those that have identical features as modified in the following paragraphs. FPT Checklist contents must include, but not be limited to, step-by-step actions, expected results, and confirmation of compliance with required provisions of ASHRAE 90.1 - IP as modified in the following paragraphs. Refer to item entitled "Construction Phase Functional Performance Testing" for more information.

3.1.10.1 Complex Systems Requirements

FPT Checklists must be prepared for all complex systems as defined in paragraph COMPLEX SYSTEMS. A single FPT Checklist must be prepared for each system except that a single FPT Checklist may be prepared for use with multiple systems that have identical control sequences such as unitary air-conditioners, packaged terminal air-conditioners, and heat pumps.

Contents must include, but not be limited to, contents specified in the following paragraphs and shown in example FPT Checklists for Section 01 91 00.15 TOTAL BUILDING COMMISSIONING available for download at the following location: <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. FPT Checklists must contain the same level of detail specified below and as shown in examples, but are not required to match the format of examples.

- a. Notable system features including information about such attributes as system sizing and controls to facilitate understanding of system operation.
- b. Conclusions and recommendations based on control system feature, point-to-point, actuator, and system operation observations. Conclusions must clearly indicate if system does or does not perform in accordance with requirements of Plans and Specifications. Recommendation must clearly indicate that the system should or should not be accepted by the Government.

- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions.
- d. Identification of the equipment involved in the test.
- e. Attendees present throughout the entire system test.
- f. Control system feature identification including control point description, embedded/visible type, adjustable/monitoring type, actual value, and setpoint value/alarm range.
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator workstation.
- h. Actuator operation observations demonstrating actuator responses to commands from the control system.
- i. Variable frequency drive observations demonstrating minimum safe motor speed.
- j. System operation observations for system-based testing demonstrating each control sequence, operation mode, and alarm condition resulting from control setpoint manipulation. System operation observations must include, but not be limited to, observations specified in the following paragraphs.
 - (1) Introduction identifying testing methodology.
 - (2) As-found conditions prior to system manipulation.
 - (3) Clear list of test items (step numbers).
 - (4) Design control sequences or interlocks segmented by unique functions.
 - (5) Intended test procedures following each segmented sequence or interlock identifying the system manipulation required to initiate system response.
 - (6) Expected system responses.
 - (7) Space for comments for each test item.
 - (8) Pass or fail indication for each test.

Performance Verification Tests prepared by the Controls Contractor as specified in Section 23 09 00.00 22 INSTRUMENTATION AND CONTROL FOR HVAC are prohibited from fulfilling requirements of this Section.

3.1.10.2 Plumbing Systems Requirements

FPT Checklists must be prepared for all equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.1.10.3 Electrical Type I Systems Requirements

FPT Checklists must not be prepared for equipment, systems, and assemblies

listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.1.10.4 Electrical Type II Systems Requirements

FPT Checklists must be prepared for all equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED. FPT Checklists must be system-based and demonstrate end-objective performance in accordance with Plans and Specifications. FPT Checklists must not repeat fundamental performance testing such as resistance measurements required in other Sections.

3.1.10.5 Building Envelope Systems, Components, and Assemblies Requirements

FPT Checklists must not be prepared for equipment, systems, and assemblies listed under item entitled "Building Envelope Systems, Components, and Assemblies Category" in paragraph SYSTEMS TO BE COMMISSIONED.

3.1.10.6 Proposed FPT Checklist Meeting and Draft FPT Checklist Meeting

For all complex systems as defined in paragraph COMPLEX SYSTEMS, a Proposed FPT Checklist Meeting and a Draft FPT Checklist Meeting must be conducted for detailed discussions of procedures intended to demonstrate performance is in accordance with Plans and Specifications. Meetings must be conducted by conference call.

3.1.10.7 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

For all equipment, systems, and assemblies other than complex systems as defined in paragraph COMPLEX SYSTEMS, the Lead and Commissioning Specialist must perform respective responsibilities specified in the following paragraphs.

- a. The Lead Commissioning Specialist must submit **Final Functional Performance Test Checklists** for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED for which FPT checklists are required to be prepared. FPT checklists must be submitted as a single submittal for each systems category no later than 30 calendar days prior to testing of any system in the respective systems category.
- b. Commissioning Specialists must prepare Final Functional Performance Test Checklists for equipment, systems, and assemblies listed under systems categories assigned to them in this Section for which FPT checklists are required to be prepared. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

For all complex systems as defined in paragraph COMPLEX SYSTEMS, the Lead Commissioning Specialist and Commissioning Specialists must prepare and submit Final Functional Performance Test Checklists in the sequence specified in the following paragraphs. Commissioning Specialists must perform respective responsibilities for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing respective responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

- a. Commissioning Specialists must prepare Proposed Functional Performance Test Checklists.
- b. The Lead Commissioning Specialist must submit **Proposed Functional Performance Test Checklists** for all complex systems as a single submittal no later than 60 calendar days prior to testing of any complex system.
- c. Commissioning Specialists must conduct the Proposed Functional Performance Test Checklist Meeting attended by the Contracting Officer's Technical Representative.
- d. Commissioning Specialists must prepare Draft Functional Performance Test Checklists to incorporate input from the Contracting Officer's Technical Representative.
- e. The Lead Commissioning Specialist must provide Draft Functional Performance Test Checklists to Installing Contractors and Equipment Suppliers no later than seven calendar days prior to the Draft Functional Performance Test Checklist Meeting.
- f. Commissioning Specialists must conduct the Draft Functional Performance Test Checklist Meeting attended by Installing Contractors and Equipment Suppliers. The Contracting Officer's Technical Representative may attend this meeting.
- g. Commissioning Specialists must prepare Final Functional Performance Test Checklists to incorporate input from Installing Contractors.
- h. The Lead Commissioning Specialist must submit Final Functional Performance Test Checklists for all complex systems as a single submittal no later than seven calendar days after meeting to discuss Draft Functional Performance Test Checklists and no later than 30 calendar days prior to testing of any complex system.

3.1.10.8 Installing Contractors and Equipment Suppliers Responsibilities

Installing Contractors and Equipment Suppliers must attend Draft Functional Performance Test Checklist Meeting.

3.1.11 Construction Submittals

3.1.11.1 General Requirements

Construction Submittals must be reviewed for all commissioned equipment, systems, and assemblies to the extent necessary to verify that equipment, systems, and assemblies are in compliance with Commissioning Aspects of Plans and Specifications as modified in this Section. One review for each commissioned equipment, system, and assembly must be performed coincident with Designers of Record and review must include, but not be limited to, review of shop drawings and equipment performance data. Review of resubmittals is prohibited.

3.1.11.2 Construction Submittal Review Reports

A Construction Submittal Review Report must be prepared for all commissioned equipment, systems, and assemblies and contents must include, but not be limited to, contents specified in the following paragraphs.

- a. Project title, project number, and date of review.
- b. Overview complete goals of the Commissioning-focused submittal review process.
- c. Summary of deficiencies.
- d. Ongoing record of all Construction Submittals received and reviewed, list of unresolved deficiencies, and record of communications made to resolve deficiencies.
- e. Descriptive outline of reviewed construction submittals complete with the Installing Contractor's submittal designation.
- f. Clear description of deficiencies complete with supporting documentation.

3.1.11.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist activities must include, but not be limited to, activities specified in the following paragraphs.

- a. Provide list of shop drawings and equipment performance data that must be reviewed to Installing Contractors.
- b. Prepare and provide Construction Submittal review comments to Designers of Record.
- c. Prepare and submit **Construction Submittal Review Reports** no later than seven calendar days after completion of each Construction Submittal review.

3.1.11.4 Commissioning Specialists Responsibilities

Commissioning Specialists must perform reviews of Construction Submittals for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.11.5 Designers of Record Responsibilities

Designers of Record activities must include, but not be limited to, receiving and considering for incorporation comments listed in Construction Submittal Review Reports provided by the Lead Commissioning Specialist.

3.1.11.6 Installing Contractors Responsibilities

Installing Contractors must provide all requested Construction Submittals and requested supplemental information to the Lead Commissioning Specialist.

3.1.12 Construction Phase Site Observations

3.1.12.1 General Requirements

Construction Phase Site Observations must be performed as specified in the following paragraphs to verify and document Commissioning Aspects of equipment, systems, and assemblies as specified in this Section. Two eight-hour Construction Phase Site Observations must be conducted for each systems category that is not specified in the following paragraphs. Specified number of observations are totals that include all commissioned buildings and all construction phases in this project.

Construction Phase Site Observations are prohibited from being combined with witnessing Functional Performance Testing, but may be combined with Pre-Functional Checklist Verification unless indicated otherwise. Purposes of Site Observations specified in the following paragraphs for each systems category may be combined into a single Site Observation.

3.1.12.2 Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. Two eight-hour Site Observations to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.
- b. One eight-hour Site Observation to verify and document HVAC pipe pressure testing.
- c. One eight-hour Site Observation to verify and document HVAC pump installation.
- d. One eight-hour Site Observation to verify and document compliance with provisions of ASHRAE 90.1 - IP.

3.1.12.3 Plumbing Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. One eight-hour Site Observation to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.
- b. One eight-hour Site Observation to verify and document Plumbing pipe pressure testing.
- c. One eight-hour Site Observation to verify and document Plumbing pump installation and compliance with provisions of ASHRAE 90.1 - IP.

3.1.12.4 Utility Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. No Site Observation dedicated to this systems category. Verification

and documentation of Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

- b. No Site Observation dedicated to this systems category to verify and document compliance with provisions of ASHRAE 90.1 - IP. Verification and documentation of compliance with provisions of ASHRAE 90.1 - IP of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

3.1.12.5 Electrical Type I Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. Two eight-hour Site Observations to verify and document Commissioning Aspects and to witness testing performed by independent testing organizations of commissioned equipment, systems, and assemblies listed under this systems category and listed by the Lead Commissioning Specialist to require on-site verification and documentation by Commissioning Specialists. Independent testing organizations typically use tests readily available from testing associations such as the InterNational Electrical Testing Association (NETA).
- b. No Site Observation dedicated to this systems category to verify and document compliance with provisions of ASHRAE 90.1 - IP. Verification and documentation of compliance with provisions of ASHRAE 90.1 - IP of commissioned equipment, systems, and assemblies listed under this systems category may be combined with Site Observations for other systems categories.

3.1.12.6 Electrical Type II Systems Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. Two eight-hour Site Observations to identify issues that may result in unsuccessful functional performance testing and to verify and document Commissioning Aspects of commissioned equipment, systems, and assemblies listed under this systems category.
- b. One eight-hour Site Observation to verify and document compliance with provisions of ASHRAE 90.1 - IP.

3.1.12.7 Building Envelope Systems, Components, and Assemblies Category

Site Observations must be performed for the purposes specified in the following paragraphs.

- a. One eight-hour Site Observation to verify, document, and witness inspections, measurements, and testing of building envelope and roofing systems performed by independent testing agencies and specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES.
- b. One eight-hour Site Observation to verify, document, and witness pressure testing, infrared thermography testing, and fog testing of

building envelope and roofing systems performed by independent testing agencies and specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.

- c. Two eight-hour Site Observation to verify and document that continuous air barrier materials and assemblies are in compliance with requirements of ASHRAE 90.1 - IP while the continuous air barrier is still accessible for inspection and repair.

3.1.12.8 Summary Meetings

A Summary Meeting must be conducted at the conclusion of each Site Observation for the purpose of discussing issues discovered during each Site Observation. Meeting must be conducted on-site with all attendees present.

3.1.12.9 Site Observation Reports

A Site Observation Report must be prepared for each Site Observation for the purpose of documenting issues discovered during each Site Observation.

Site Observation Reports for all equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include results of testing performed by independent testing organizations that were witnessed by Commissioning Specialists and test results that were not witnessed by Commissioning Specialists.

3.1.12.10 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must provide list of equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED that require on-site verification and documentation by Commissioning Specialists to Installing Contractors.

The Lead Commissioning Specialist must submit **Construction Phase Site Observation Reports** complete with Summary Meeting Attendance Roster no later than seven calendar days after Site Observation.

3.1.12.11 Commissioning Specialists Responsibilities

Commissioning Specialist must perform responsibilities specified in the following paragraphs.

- a. Each Commissioning Specialist must be responsible to perform Site Observations, conduct Summary Meetings, and prepare Construction Phase Site Observation Reports for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.12.12 Installing Contractors Responsibilities

Installing Contractors must provide Commissioning Specialists access to construction and systems for Site Observations and must attend Summary Meetings.

Installing Contractors must schedule testing to occur in the presence of Commissioning Specialists as specified in the following paragraphs.

- a. All equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED and listed by the Lead Commissioning Specialist to require on-site verification and documentation by Commissioning Specialists. Results of testing of equipment, systems, and assemblies listed under item entitled "Electrical Type I Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must be provided to the Lead Commissioning Specialist that were witnessed by Commissioning Specialists and that were not witnessed by Commissioning Specialists.
- b. Pressure testing, infrared thermography testing, and fog testing specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING and inspections, measurements, and testing specified in Sections 07 42 13 METAL WALL PANELS and 07 42 63 METAL WALL PANEL ASSEMBLIES for equipment, systems, and assemblies listed under item entitled "Building Envelope Systems, Components, and Assemblies Category" in paragraph SYSTEMS TO BE COMMISSIONED. Results of all tests, inspections, and measurements must be provided to the Lead Commissioning Specialist that were witnessed by Commissioning Specialists and that were not witnessed by Commissioning Specialists.

3.1.13 Pre-Functional Checks

Pre-Functional Checks must be performed prior to performing Functional Performance Testing of respective system and activities must include, but not be limited to, activities specified in the following paragraphs.

- a. Perform Pre-Functional Checklist Verification.

3.1.13.1 Pre-Functional Checklist Verification

3.1.13.1.1 General Requirements

Pre-Functional Checklist Verification must be performed using Pre-Functional Checklists from the approved Construction Phase Final Commissioning Plan by conducting on-site spot-checks to verify accuracy after receipt of completed Pre-Functional Checklists from Installing Contractors. Verification may be combined with Construction Phase Site Observations.

3.1.13.1.2 Number of Trips

The number of trips to the site for the purpose of Pre-Functional Checklist Verification for each systems category must be up to that specified in the following paragraphs. It is encouraged that these trips be combined with Construction Phase Site Observations.

- a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: Two eight-hour trips.
- b. Plumbing Systems Category: Two eight-hour trips.
- c. Utility Systems Category: No trip dedicated to this systems category. Verification of commissioned equipment, systems, and assemblies listed under this systems category may be combined with trips for other systems categories.

- d. Electrical Type I Systems Category: One eight-hour trip.
- e. Electrical Type II Systems Category: Two eight-hour trips.
- f. Building Envelope Systems, Components, and Assemblies Category: One eight-hour trip.

3.1.13.1.3 Sample Strategy

20-Percent sample spot-check for all equipment, systems, and assemblies for which Pre-Functional Checklists have been completed. Sample spot-check must include at least one complete item such as one complete unitary air-conditioner.

3.1.13.1.4 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must provide blank Pre-Functional Checklists to Installing Contractors.

The Lead Commissioning Specialist must submit all **completed Pre-Functional Checklists** and all associated equipment start-up reports for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than seven calendar days after receipt of all completed checklists for equipment, systems, and assemblies listed under the respective systems category from Installing Contractors.

3.1.13.1.5 Commissioning Specialists Responsibilities

Commissioning Specialists must perform Pre-Functional Checklist Verifications and then must indicate acceptance of all Pre-Functional Checklist items for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.13.1.6 Installing Contractors Responsibilities

Installing Contractors must complete one Pre-Functional Checklist for each piece of equipment and system. Initial each Pre-Functional Checklist item for their area of responsibility indicating the item's construction, installation, and start-up as applicable has been verified to be complete and is ready for testing.

Installing Contractors must provide completed and initialed Pre-Functional Checklists and all associated equipment start-up reports to the Lead Commissioning Specialist as a single package for each systems category no later than seven calendar days after completion of inspection of all checklist items for each piece of equipment and system.

3.1.14 Certified Certificates of Readiness and Certificates of Readiness

3.1.14.1 General Requirements

A single Certified Certificate of Readiness must be prepared for each equipment, system, and assembly subject to Functional Performance Testing and testing required according to UFGS 01 91 19 BUILDING ENCLOSURE COMMISSIONING by reviewing, signing, and dating Certificates of Readiness

to indicate each equipment, system, and assembly is complete and ready for testing. Testing of any equipment, system, or assembly is prohibited until after its Certified Certificate of Readiness has been submitted and approved. Government approval of Certified Certificates of Readiness does not constitute the Government's responsibility for equipment, systems, or assemblies being complete and ready for testing.

A single Certificate of Readiness must be prepared for each equipment, system, and assembly subject to Functional Performance Testing. Certificate of Readiness contents must include, but not be limited to, contents specified in the following paragraphs to the extent applicable to the specific system being tested.

- a. Cover sheet with spaces for signatures and dates by the Lead Commissioning Specialist and by Installing Contractors to indicate each system is complete and ready for testing.
- b. Equipment and system start-up reports.
- c. Control system start-up reports.
- d. Diagnostic test reports.
- e. Completed Pre-Functional Checklists.

3.1.14.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must sign, date, and submit **Certified Certificates of Readiness** for all equipment, systems, and assemblies listed under each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days prior to testing of any equipment, system, or assembly listed under respective systems category.

3.1.14.3 Commissioning Specialists Responsibilities

Commissioning Specialists must perform reviews of Certificates of Readiness for accuracy to ensure equipment, systems, and assemblies are complete and ready for testing for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.14.4 Installing Contractors Responsibilities

Installing Contractors must prepare, sign, and date Certificates of Readiness indicating equipment, systems, and assemblies are complete and ready for testing. Certificates of Readiness for all equipment, systems, and assemblies listed under each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED must be provided to the Lead Commissioning Specialist as a single package for each systems category.

3.1.15 Construction Phase Functional Performance Testing

3.1.15.1 General Requirements

Construction Phase Functional Performance Testing must be performed for all equipment, systems, and assemblies for which a Functional Performance

Test (FPT) Checklist has been prepared. Testing must be performed using approved checklists and all required Commissioning Team participants must be present prior to start of testing. Testing must be aborted and rescheduled to occur at a later date if any Commissioning Team participant is not present or if deficiencies discovered during testing prevent successful completion. Testing of any equipment, system, and assembly is prohibited until after its Certified Certificate of Readiness has been submitted and approved.

All necessary materials and system modifications to simulate actual conditions as close as is practically possible must be provided. Complete all testing by returning affected equipment, systems, and assemblies to pre-test conditions.

3.1.15.2 Complex Systems Requirements

Testing of complex systems as defined in paragraph COMPLEX SYSTEMS in paragraph SYSTEMS TO BE COMMISSIONED must be performed as specified in the following paragraphs.

3.1.15.2.1 Equipment Featuring On-Board Factory Packaged Controls

Testing of equipment featuring on-board factory packaged controls must include, but not be limited to, testing specified in the following paragraphs.

- a. Equipment enabling and disabling.
- b. Equipment standard and optional control points necessary to accomplish functionality whether or not control points are specified in Plans and Specifications.
- c. Equipment standard and optional alarms critical to safe operation whether or not optional alarms are specified in Plans and Specifications.
- d. All control points added by Controls Contractor in addition to on-board factory packaged controls whether or not control points are specified in Plans and Specifications.

3.1.15.2.2 Test Setup and Condition Simulation

Test setup must include all necessary materials and system modifications to simulate flows, pressures, temperatures, and other conditions necessary to perform testing according to specified conditions must be provided.

Condition simulation must be accomplished by temporarily changing control system setpoints. Condition simulation by over-writing control system inputs (actual) values is prohibited unless approved in advance of test by the Contracting Officer's Technical Representative.

3.1.15.2.3 Condition Simulation Methods

Condition simulation methods for equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED that are acceptable are specified in the following paragraphs.

- a. When varying static pressures inside ductwork cannot be simulated

within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various setpoints, nor to simulate an electric analog signal unless approved as specified in the preceding paragraphs.

- b. Dirty filter pressure drops can be simulated using sheets of cardboard at filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. High entering cooling coil temperatures can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer's Technical Representative, as noted above, for special cases.
- g. Control setpoints can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below lockout setpoint, when the outside air temperature is above lockout setpoint, temporarily change the lockout setpoint to be minus 0 degrees above the current outside air temperature. Caution: Setpoints are not to be raised or lowered to a point such that damage to the components, systems, or the building structure and/or contents will occur.
- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform testing with air system at minimum airflow condition in ductwork.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the setpoint on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is performed, return the setpoint to its original setpoint or a setpoint as indicated by the Contracting Officer's Technical Representative.

3.1.15.3 Sample Strategy

3.1.15.3.1 General Requirements

100-Percent testing of all equipment, systems, and assemblies as modified in the following paragraphs. Sample strategy for testing less than 100-percent testing must be equally applied to each commissioned building in this project.

Sample strategy testing must be applied to total equipment and system counts specified for system categories under paragraph SYSTEMS TO BE COMMISSIONED.

3.1.15.3.2 Complex Systems

Sample strategy for all complex systems as defined in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

- a. 100-Percent testing for equipment, systems, and assemblies such as chilled water systems, HVAC heating hot water systems, renewable energy systems, building automation systems, energy management and control systems, aircraft pre-conditioned air systems, air handling unit systems including all associated fans except for remote exhaust air fans, dedicated outside air systems including all associated fans except for remote exhaust air fans, computer room air-conditioning units, whole-building energy monitoring, and whole-building water utility monitoring.
- b. 20-Percent sample testing for equipment, systems, and assemblies with identical control sequences such as heat pumps, split system air-conditioning units, and packaged terminal air-conditioner units. Sample must include at least one complete item such as one complete heat pump.

3.1.15.3.3 Plumbing Systems

Sample strategy for all equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED except plumbing systems defined as complex systems in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

- a. 100-Percent testing for equipment, systems, and assemblies such as domestic heating hot water systems, domestic solar heating hot water systems, service water pressure-booster pump systems, process water systems, natural gas systems, and irrigation systems.
- b. 20-Percent sample testing for equipment, systems, and assemblies with identical features such as fixtures, semi-instantaneous water heaters, emergency eyewash stations, and vacuum systems. Sample must include at least one complete item such as one complete fixture.

3.1.15.3.4 Electrical Type II Systems

Sample strategy for all equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED except electrical systems defined as complex systems in paragraph COMPLEX SYSTEMS must be as specified in the following paragraphs.

- a. 100-Percent testing for equipment, systems, and assemblies such as frequency converters, emergency power off control systems, emergency response systems, electrical utility metering systems, and photovoltaic systems.
- b. 20-Percent sample testing for equipment, systems, and assemblies with identical features such as audiovisual systems, automatic receptacles controls, communications systems, energy monitoring, lighting systems, lighting control devices, lighting control systems, and security systems. Sample must include at least one complete item such as lighting in one space.

3.1.15.4 Unsuccessful Testing and Re-Testing

3.1.15.4.1 Unsuccessful Testing of 100-Percent Sample

Testing must be failed for equipment or systems for which 100-percent sample size are tested if one or more of the test procedures results in discovery of a deficiency that cannot be resolved within five minutes during testing.

Re-testing must be witnessed to the extent necessary to confirm that deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.1.15.4.2 Unsuccessful Testing of Less than 100-Percent Sample

Testing must be failed for equipment or systems for which less than 100-percent sample size are tested if one or more of the test procedures results in discovery of a deficiency, regardless of whether the deficiency is corrected during the sample test.

- a. Re-testing must be witnessed of only items which experienced initial failures if the failure rate is 5-percent or less, meaning that 5-percent or less of the equipment or systems had at least one deficiency.
- b. Re-testing must be witnessed of items which experienced initial failures to the extent necessary to confirm that the deficiencies have been corrected if the failure rate is higher than 5-percent, meaning that more than 5-percent of equipment or systems tested had at least one deficiency. In addition, testing of another random sample must be witnessed for the same size as the initial sample for the first time. If the second random sample set has any failures, those failed items must be retested and all remaining equipment, systems, and assemblies must be tested to complete 100-percent testing of that system type.

3.1.15.5 Re-Testing Costs

Re-testing costs may be withheld by the Contracting Officer from payments to Installing Contractors. These costs may include salary, travel costs, and per diem associated with re-testing for the Lead Commissioning Specialist, Commissioning Specialists, and Government personnel.

3.1.15.6 Commissioning Team Participants

Participants of Functional Performance Testing must include, but not be limited to, participants specified in the following paragraphs. The Contracting Officer's Technical Representative, Public Works Department Division (PWD) Representatives, and Using Agent's Representatives (Users) may, but are not required to, participate in any testing.

3.1.15.6.1 Heating, Ventilation, and Air-Conditioning (HVAC) Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

- a. Mechanical Commissioning Specialist (CxM).

- b. Installing Contractors' Quality Control Personnel (CQC).
- c. Installing Contractors' Controls Commissioning Representative (CC).
- d. Installing Contractors' TAB Commissioning Representative (TAB).
- e. Installing Contractors' Mechanical Commissioning Representative (MC).
- f. Equipment Supplier (ES).

3.1.15.6.2 Plumbing Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Plumbing Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

- a. Mechanical Commissioning Specialist (CxM).
- b. Installing Contractors' Quality Control Personnel (CQC).
- c. Installing Contractors' Controls Commissioning Representative (CC).
- d. Installing Contractors' Plumbing Commissioning Representative (PC).
- e. Equipment Supplier (ES).

3.1.15.6.3 Utility Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Utility Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

- a. Mechanical Commissioning Specialist (CxM).
- b. Installing Contractors' Quality Control Personnel (CQC).
- c. Installing Contractors' Controls Commissioning Representative (CC).
- d. Installing Contractors' TAB Commissioning Representative (TAB).
- e. Installing Contractors' Mechanical Commissioning Representative (MC).
- f. Installing Contractors' Plumbing Commissioning Representative (PC).
- g. Installing Contractors' Electrical Commissioning Representative (EC).
- h. Equipment Supplier (ES).

3.1.15.6.4 Electrical Type II Systems

Commissioning Team members participating in Functional Performance Testing of equipment, systems, and assemblies listed under item entitled "Electrical Type II Systems Category" in paragraph SYSTEMS TO BE COMMISSIONED must include, but not be limited to, participants specified in the following paragraphs.

- a. Electrical Commissioning Specialist (CxE).
- b. Installing Contractors' Quality Control Personnel (CQC).
- c. Installing Contractors' Electrical Commissioning Representative (EC).
- d. Equipment Supplier (ES).

3.1.15.7 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

- a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to performing Testing to select which systems must be tested for samples of less than 100-percent.
- b. Coordinate logistics with Installing Contractors.
- c. Submit **completed Final Functional Performance Test Checklists** for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days after successful testing of all systems listed under the respective systems category.

3.1.15.8 Commissioning Specialists Responsibilities

Commissioning Specialists must witness, re-witness, and document all testing for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.15.9 Installing Contractors Responsibilities

Installing Contractors must provide all necessary materials and perform all aspects of testing including modifications to simulate actual conditions for all equipment, systems, and assemblies related to their contract.

3.1.15.10 Equipment Suppliers Responsibilities

Equipment Suppliers must perform all aspects of testing of equipment featuring on-board factory packaged controls provided as part of their contract. Refer to paragraph COMPLEX SYSTEMS for definition of equipment featuring on-board factory packaged controls.

3.1.15.11 Installing Contractors and Equipment Supplier Coordination

Installing Contractors and Equipment Suppliers must simultaneously perform all aspects of testing of equipment featuring on-board factory packaged controls provided as part of their contract.

3.1.16 Training

3.1.16.1 Training Plans

Training Plans must be prepared for all commissioned equipment, systems, and assemblies and contents must include, but not be limited to, contents

specified in the following paragraphs.

- a. List of those who should receive training by position or name.
- b. List of systems that require training organized by Specification Section.
- c. Outline of training topics that addresses design, construction, operation, and maintenance.
- d. Learning objectives and training delivery methods.
- e. Planned location of the training sessions (classroom, on-site, and off-site).
- f. Minimum duration of each training session.
- g. Training materials requirements to be used during the instructional process.
- h. Training report, records, and recording requirements.
- i. Instructor qualifications.
- j. Instructor name and contact information.
- k. Criteria for determining whether provided training is acceptable.
- l. Blank Training Attendance Rosters.

3.1.16.2 Training

Training must be performed in accordance with Training Plans and in accordance with requirements specified in other Sections.

3.1.16.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, provide all review comments to the Installing Contractors for all equipment, systems, and assemblies listed under each systems category as a single package no later than 45 calendar days prior to first training event for any equipment, system, or assembly in respective systems category. Review comments are not required to be submitted to the Government.

3.1.16.4 Commissioning Specialist Responsibilities

Commissioning Specialists must, at a minimum, review and prepare review comments for all Training Plans. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

3.1.16.5 Installing Contractors Responsibilities

Installing Contractors must, at a minimum, perform activities specified in the following paragraphs.

- a. Prepare and provide all Training Plans to the Lead Commissioning Specialist for all equipment, systems, and assemblies listed under

each systems category as a single package no later than 60 calendar days prior to first training event for any equipment, system, or assembly in respective systems category.

- b. Incorporate all review comments.
- c. Perform training as specified in the preceding paragraphs.
- d. Require all training attendees to sign Training Attendance Rosters.
- e. Provide all Approved Training Plans and Completed Training Attendance Rosters to the Lead Commissioning Specialist for all equipment, systems, and assemblies listed under each systems category as a single package no later than 30 calendar days after the last training event for all equipment, systems, and assemblies in respective systems category.

3.1.17 Final Commissioning Report

3.1.17.1 General Requirements

The Final Commissioning Report must be prepared and contents must include, but not be limited to, contents specified in the following paragraphs.

- a. Executive summary describing the overall Commissioning Process.
- b. Results of the Commissioning Process.
- c. Issues And Resolution Log.
- d. Commissioning Design Review Reports.
- e. Approved Construction Phase Final Commissioning Plan.
- f. Approved Pre-Final Test and Balance Reports.
- g. Completed Pre-Functional Checklists.
- h. Completed Functional Performance Test Checklists.
- i. Approved Training Plans.
- j. Completed Training Attendance Rosters.
- k. Indication that equipment, systems, and assemblies meet requirements of the Owner's Project Requirements, Plans and Specifications, and provisions of ASHRAE 90.1 - IP.
- l. Indication of plan for completion of any deferred trainings.
- m. Identification of all deficiencies complete with corrective actions taken.
- n. Identification of all repairs, replacements, and calibrations.
- o. Identification of unresolved deficiencies complete with resolution plan and responsible Commissioning Team member.
- p. List and plan for completion of any testing that must be performed

during the Post-Construction (Warranty) Phase.

3.1.17.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

- a. Prepare and submit the **Final Commissioning Report** no later than 30 calendar days after Beneficial Occupancy.

3.2 POST-CONSTRUCTION (WARRANTY) PHASE

3.2.1 Post-Construction Phase ASHRAE Standard 90.1 Commissioning Compliance

3.2.1.1 General Requirements

Activities must include, but not be limited to, Post-Construction Phase Commissioning Activities referred to as "Building Systems Verification and Testing" and "Building Commissioning" in ASHRAE 90.1 - IP as modified in this Section, Post-Construction Phase Commissioning Activities listed in references contained in ASHRAE 90.1 - IP as modified in this Section, and Post-Construction Phase Commissioning Activities specified in this Section.

3.2.1.2 Lead Commissioning Specialist and Commissioning Specialists Responsibilities

The Lead Commissioning Specialist and Commissioning Specialists must, at a minimum, perform all activities specified under item entitled "Post-Construction Phase ASHRAE Standard 90.1 Commissioning Compliance."

3.2.2 Post-Construction Phase Issues And Resolution Log

3.2.2.1 General Requirements

The Post-Construction Phase Issues and Resolution Log must be prepared and maintained to track all Commissioning Issues identified by any member of the Commissioning Team. The Post-Construction Phase Issues and Resolution Log must be prepared in an electronic format that can be used throughout the Post-Construction Phase such that it can be provided to the Government upon request. Post-Construction Phase Issues and Resolution Log contents must include, but not be limited to, contents specified in the following paragraphs.

- a. Project title and location.
- b. Lead Commissioning Specialist's contact information.
- c. Issue number.
- d. Date issue was discovered.
- e. Name of Commissioning Team member identifying the issue.
- f. Commissioning Team member responsible for resolving the issue.
- g. Current status.
- h. Issue description.

- i. Pictures of item if available and appropriate.
- j. Effects of issue on project or building operation.
- k. Possible cause of issue.
- l. Proposed resolution.
- m. Final resolution and final resolution date.
- n. Approvals of issue resolution, including approver's name.

3.2.2.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must prepare and maintain the **Post Construction Phase Issues And Resolution Log** and must submit one electronic copy no later than 14 calendar days after completion of any Post-Construction Phase Commissioning Activity.

3.2.3 Post-Construction Phase Site Observations

3.2.3.1 General Requirements

Post-Construction Phase Site Observations must be performed for each systems category specified in the following item entitled "Number of Trips" between eight and 10 months after Beneficial Occupancy to verify and document that all commissioned equipment, systems, and assemblies are operating in accordance with the Plans and Specifications.

Post-Construction Phase Site Observations are prohibited from being combined with witnessing Post-Construction Phase Functional Performance Testing.

3.2.3.2 Number of Trips

The Number of Post-Construction Phase Site Observations inclusive of all commissioned buildings in this project must be as follows

- a. Heating, Ventilation, and Air-Conditioning (HVAC) Systems Category: One eight-hour trip.
- b. Plumbing Systems Category: No trips. Combine these activities with HVAC Systems Category activities.
- c. Utility Systems Category: No trips.
- d. Electrical Type I Systems Category: No trips.
- e. Electrical Type II Systems Category: One eight-hour trip.
- f. Building Envelope Systems, Components, and Assemblies Category: One eight-hour trip.

3.2.3.3 Site Observation Reports

Site Observation Reports must be prepared for each Site Observation for the purpose of documenting issues discovered during Site Observation.

3.2.3.4 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

- a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to performing Site Observation to schedule attendance of Operations and Maintenance staff and occupants and to make provisions for access to all commissioned equipment, systems, and assemblies.
- b. Submit **Post-Construction Phase Site Observation Reports** no later than 14 calendar days after Site Observation.

3.2.3.5 Commissioning Specialists Responsibilities

Commissioning Specialists must, at a minimum, perform activities specified in the following paragraphs for equipment, systems, and assemblies listed under systems categories assigned to them in this Section. Commissioning Specialists are prohibited from performing these responsibilities for equipment, systems, and assemblies listed under systems categories assigned to other Commissioning Specialists.

- a. Interview Operations and Maintenance staff.
- b. Interview occupants.
- c. Determine status of unresolved Commissioning Issues.
- d. Compare current operations to operations identified throughout the Commissioning Process.
- e. Prepare Post-Construction Phase Site Observation Reports.

3.2.4 Certified Certificates of Readiness and Certificates of Readiness

Certified Certificates of Readiness and Certificates of Readiness are not required for any Post-Construction Phase activity.

3.2.5 Post-Construction Phase Functional Performance Testing

3.2.5.1 General Requirements

Post-Construction Phase Functional Performance Testing must be performed for all commissioned equipment, systems, and assemblies that could not be fully or successfully tested during the Construction Phase. Testing must be performed using approved checklists and all required Commissioning Team participants must be present prior to start of testing. Testing must be aborted and rescheduled to occur at a later date if any Commissioning Team participant is not present or if deficiencies discovered during testing prevent successful completion.

Post-Construction Phase Functional Performance Testing requirements and activities must include, but not be limited to, requirements and activities specified for Construction Phase Functional Performance Testing as modified in paragraphs under paragraph POST-CONSTRUCTION (WARRANTY) PHASE.

3.2.5.2 Commissioning Team Participants

Participants of Functional Performance Testing must include, but not be limited to, participants specified in the following paragraphs. The Contracting Officer's Technical Representative, Public Works Department Division (PWD) Representatives, and Using Agent's Representatives (Users) may, but are not required to, participate in any testing.

- a. Lead Commissioning Specialist.
- b. Commissioning Specialists.
- c. Installing Contractors.
- d. Equipment Suppliers.

3.2.5.3 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, perform activities specified in the following paragraphs.

- a. Contact Contracting Officer's Technical Representative no later than 30 calendar days prior to witnessing Post-Construction Phase Functional Performance Testing to make provisions for access to all commissioned equipment, systems, and assemblies.
- b. Coordinate logistics with Installing Contractors.
- c. Submit completed Final Functional Performance Test Checklists for each systems category listed under paragraph SYSTEMS TO BE COMMISSIONED as a single submittal no later than 14 calendar days after successful testing of all systems listed under the respective systems category.

3.2.6 Updated Final Commissioning Report

3.2.6.1 General Requirements

The Updated Final Commissioning Report must be prepared and must be an update to the Final Commissioning Report prepared during the Construction Phase and contents must include, but not be limited to, changes resulting from Post-Construction Phase Commissioning Activities.

3.2.6.2 Lead Commissioning Specialist Responsibilities

The Lead Commissioning Specialist must, at a minimum, prepare and submit the **Updated Final Commissioning Report** no later than 30 calendar days after completion of all Functional Performance Testing.

-- End of Section --